



SMART AGRO AI

**FROM SOIL TO MARKET : INTELLIGENT DECISIONS
AT EVERY STAGE OF FARMING**

TEAM NAME : LAKSHMI BOYZZ

YASH PATIL

TANMAY RAHANE

TEJAS GHOTI

SAMEER DHANDE

YASH AMANE

CybageKhushaboo

Tech Fest

2026



PROBLEM STATEMENT

Key Challenges Faced by Farmers

- No accurate soil & nutrient analysis
- Wrong crop selection → low yield
- Late disease detection → crop loss
- Unpredictable mandi prices → low profit
- Disconnected tools for each task
- High cost of smart farming solutions
- No knowledge of Government schemes.

Need of the Hour

A single, affordable, farmer-friendly web platform that can:

- Recommend what to grow
- Predict yield & diseases early
- Advise when to sell for maximum profit
- Guide for schemes and compensation .

Core Gap

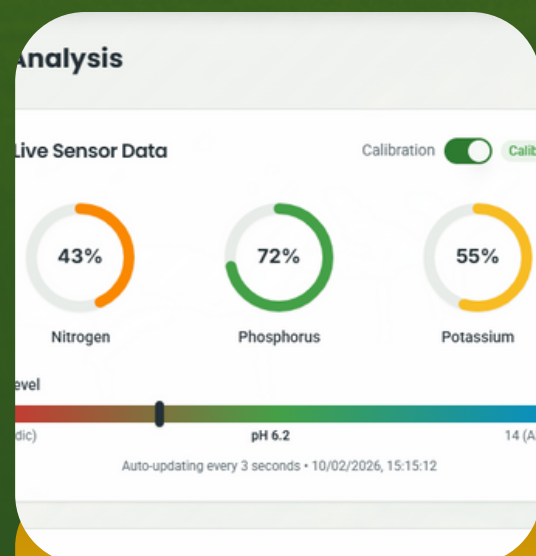
Data is available, but decision intelligence is missing.



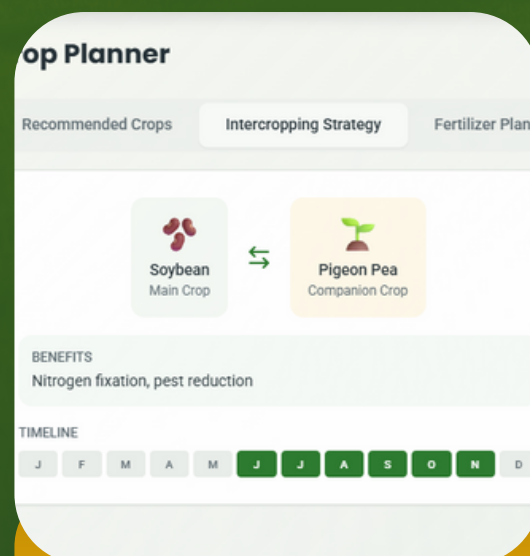


OUR SOLUTION — SMART AGRO AI

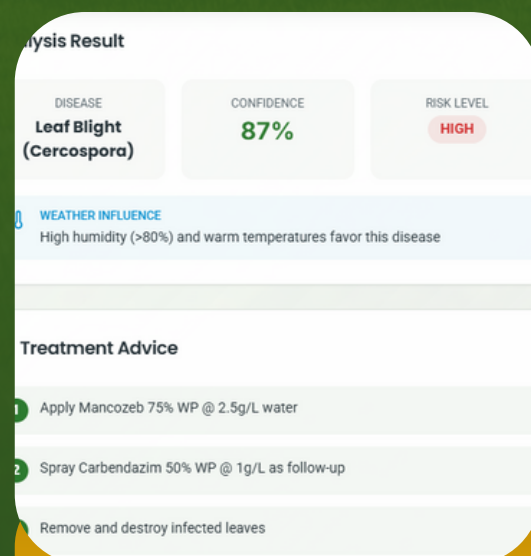
SMART AGRO AI IS A CLOSED-LOOP, AI-DRIVEN SMART AGRICULTURE PLATFORM THAT UNIFIES SOIL INTELLIGENCE, CROP PLANNING, DISEASE DETECTION, YIELD FORECASTING, MARKET ADVISORY, AND GOVERNMENT SCHEME AWARENESS INTO A SINGLE, FARMER-FRIENDLY WEB SYSTEM.



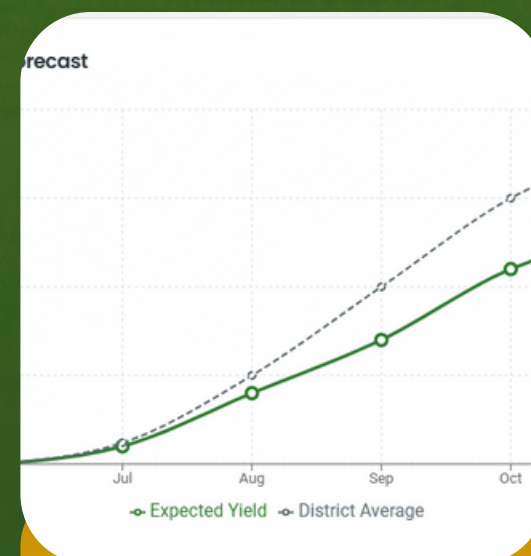
SOIL ANALYSIS



CROP PLANNER



DISEASE
DETECTION



YEILD FORECAST

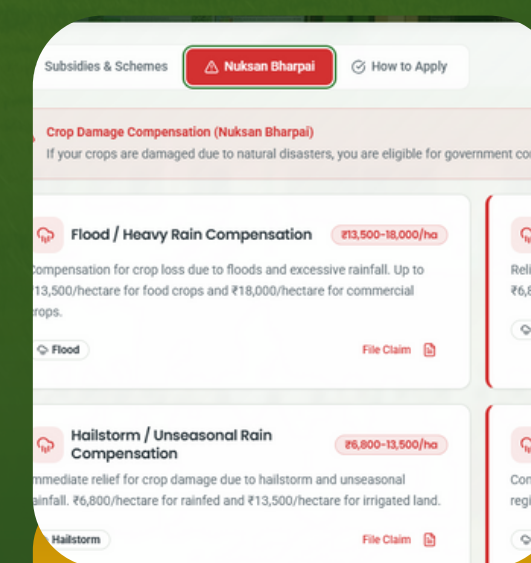
Market Advisor

Select Crop: Soybean

Select Mandi: Nagpur

Date	Predicted Price	Confidence	Recommendation
2026-02-10	₹4920	82%	HOLD
2026-02-15	₹5050	75%	HOLD
2026-02-20	₹5180	68%	SELL
2026-02-25	₹5100	61%	HOLD
2026-03-01	₹4980	55%	SELL

MARKET
ADVISER



GOVERNMENT
SCHEMES





SYSTEM ARCHITECTURE & WORKFLOW

Input Layer

- Soil Data (NPK, pH)
- Weather Data (API)
- Crop Images
- Market Prices
- Farmer Profile

AI & Analytics Layer

- Soil Intelligence Engine
- Disease Detection (Image AI)
- Yield Forecasting (ML Models)
- Market Advisory (Price Prediction)

Decision Engine (Closed Loop)

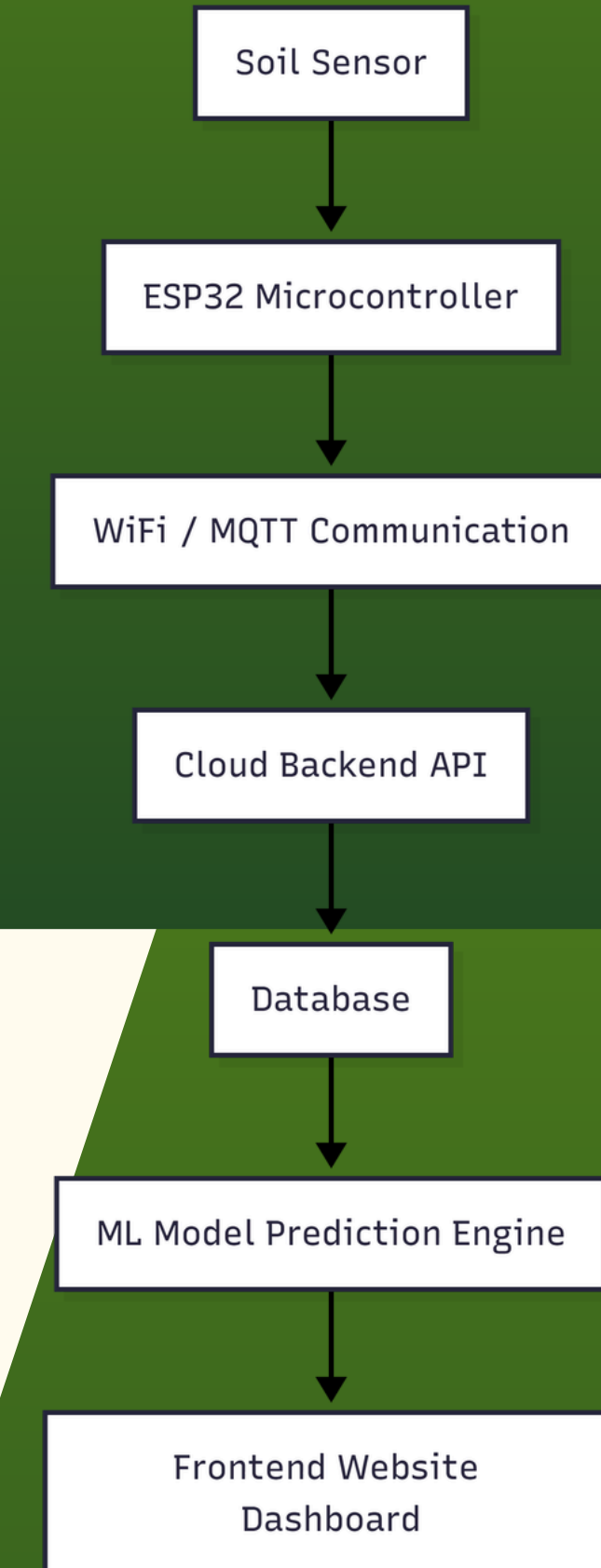
- Data Fusion & Risk Analysis
- Actionable Recommendations
- Continuous Learning

Application Layer

- Web Dashboard (EN / HI)
- Charts, Alerts, Reports
- Low-End Device Optimized

Output Layer

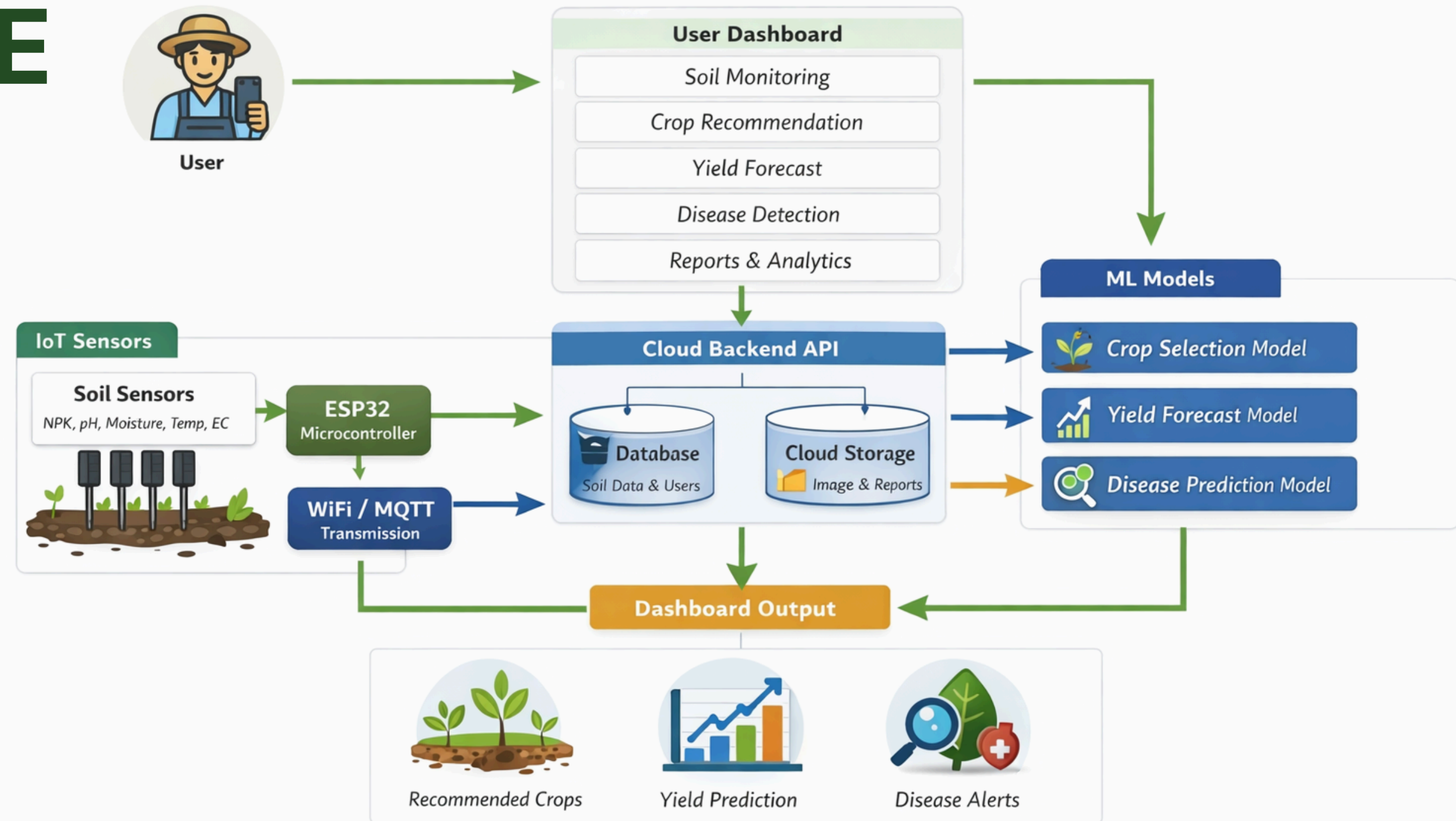
- Crop & Fertilizer Advice
- Disease Alerts
- Yield Estimation
- Sell / Hold Recommendation
- Govt Schemes & Nuksan Bharpai Info





COMPLETE ARCHITECTURE

Smart Agro AI – Complete System Overview





SOIL & CROP INTELLIGENCE

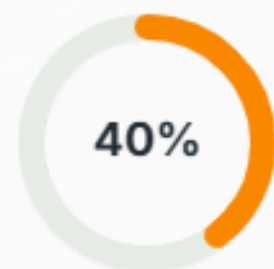


The system analyzes soil nutrients (N, P, K), soil pH, and local weather conditions using a season- and location-aware multi-label AI model. It generates a primary crop recommendation along with a compatible intercrop and a suitability score. Domain knowledge such as nutrient balance, root depth compatibility, and water requirement alignment is applied to ensure agronomically valid decisions.

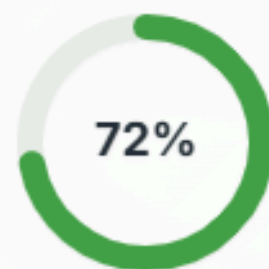
Soil Analysis

Live Sensor Data

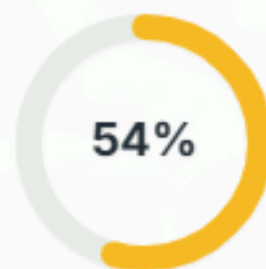
Calibration ☒ Calibrated



Nitrogen

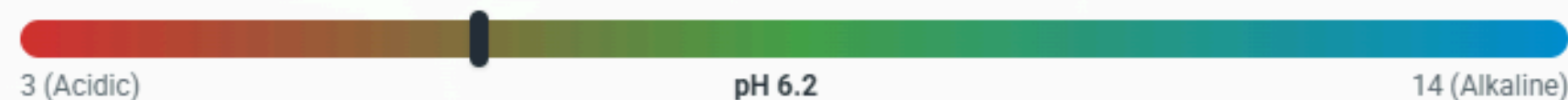


Phosphorus



Potassium

pH Level



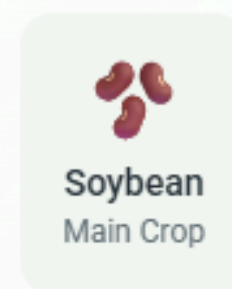
Auto-updating every 3 seconds • 10/02/2026, 15:49:58

Crop Planner

Recommended Crops

Intercropping Strategy

Fertilizer Plan



Soybean
Main Crop



Pigeon Pea
Companion Crop

BENEFITS

Nitrogen fixation, pest reduction

TIMELINE





DISEASE & LOSS PREDICTION



The application uses image-based AI to detect crop diseases from leaf or plant images uploaded by farmers. A lightweight vision model analyzes visual patterns and combines them with contextual weather data such as humidity and rainfall to improve accuracy. The system estimates disease risk, predicts potential yield loss, and provides early warnings to enable timely preventive action.

Disease Detection

Upload Crop Image



Image uploaded!
soybean_leaf_01.jpg

Analysis Result

DISEASE	CONFIDENCE	RISK LEVEL
Leaf Blight (Cercospora)	87%	HIGH

WEATHER INFLUENCE

High humidity (>80%) and warm temperatures favor this disease

Treatment Advice

- 1 Apply Mancozeb 75% WP @ 2.5g/L water
- 2 Spray Carbendazim 50% WP @ 1g/L as follow-up
- 3 Remove and destroy infected leaves

SAFETY NOTES

Wear mask and gloves during spraying. Do not spray during windy conditions. Wait 7 days before harvest.



GOVERNMENT SCHEMES & RISK SUPPORT

The platform integrates information from major government agricultural schemes and insurance programs to support farmer risk management. Based on crop type, season, and location, it highlights applicable schemes such as crop insurance, disaster compensation, and subsidy programs. In case of predicted yield loss or disease risk, the system guides farmers toward relevant nuksan bharpai and claim procedures.

₹ Subsidies & Schemes

⚠ Nuksan Bharpai

✓ How to Apply

All

Income Support

Insurance

Credit

Equipment

Irrigation

₹ PM-KISAN Samman Nidhi

₹6,000/yr

₹6,000/year direct income support to all land-holding farmer families in three equal installments.

✓ Active

Apply Now →

✓ Pradhan Mantri Fasal Bima Yojana (PMFBY)

2% Premium

Crop insurance at just 2% premium for Kharif and 1.5% for Rabi crops. Covers natural calamities, pests and diseases.

₹ Subsidies & Schemes

⚠ Nuksan Bharpai

✓ How to Apply

⚠ Crop Damage Compensation (Nuksan Bharpai)

If your crops are damaged due to natural disasters, you are eligible for compensation.

🌊 Flood / Heavy Rain Compensation

₹13,500-18,000/ha

Compensation for crop loss due to floods and excessive rainfall. Up to ₹13,500/hectare for food crops and ₹18,000/hectare for commercial crops.

🌊 Flood

File Claim 📄

🌊 Hailstorm / Unseasonal Rain

₹6,800-13,500/ha

📄 How to Apply for Schemes

1 Visit CSC / Gram Panchayat Office

Go to your nearest Common Service Centre or Gram Panchayat office with all documents.

2 Fill Application Form

Fill in the application form with your Aadhaar, bank details, and land records.

3 Submit Documents

Submit photocopies of all required documents. Keep originals for verification.

4 Track Application Status

Track your application online at the official portal or call the helpline number.

📄 Document Checklist

✓ Aadhaar Card

✓ Land Ownership Document

✓ Bank Passbook

✓ Caste Certificate

✓ Crop Sowing Date

✓ Passport Size Photo



YIELD FORECASTING



The platform predicts crop yield by analyzing historical production data, seasonal weather patterns, and current field conditions. A time-series based AI model captures long-term climate impact and growth trends to estimate expected yield before harvest. This helps farmers plan storage, labor, and financial decisions with greater confidence.

Weather Scenario

Good Weather

Average Weather

Poor Weather

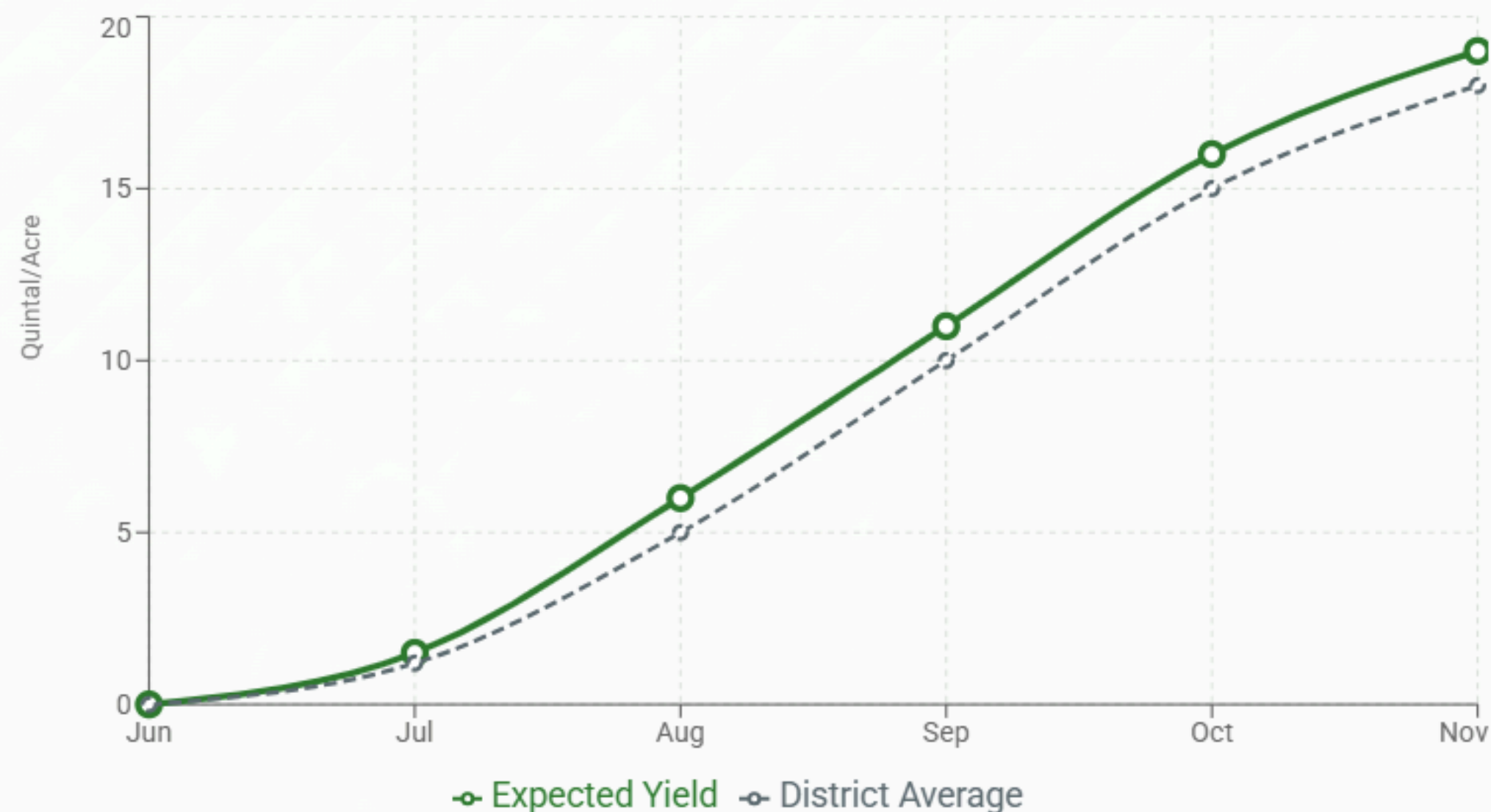
Expected Yield

19

Quintal/Acre

↑ +2.0 vs Last Season (17)

Yield Forecast





COST ESTIMATION

Cloud Cost Breakdown (AWS / Azure / GCP)

• Component	Monthly Cost (₹)
• Compute (API + ML Models)	0.01
• Database Operations	0.01
• Database Storage	0.01
• Image Storage	0.02
• Bandwidth (Data Transfer)	0.10 – 0.15
• Total Cost per User / Month	₹0.15 – ₹0.25

Scalability Example

- 1,000 users → ₹150 – ₹250 / month
- 10,000 users → ₹1,500 – ₹2,500 / month





COST ESTIMATION

Cloud Cost Per User (Dynamic Usage Model)

Usage Scenario (Per User)

- Soil sensor active 10 minutes per day
- 10 sensor uploads per day (300/month)
- 3 ML models used daily:
 - Yield Forecast
 - Crop Selection
 - Disease Prediction
- 5 website visits per day
- 15 image uploads per month (avg.)



🟡 VERSION 2: Advanced Production Version

For 10,000+ users.

👤 Developer Infrastructure

Component	Cloud	Monthly Cost	📄
Backend (2 instances)	AWS EC2	₹6000	
Load Balancer	AWS ALB	₹1500	
Database	RDS	₹4000	
Storage	S3	₹2000	
CDN	Cloudflare	₹1000	
Monitoring	Grafana / CloudWatch	₹1000	

Total:

👉 ₹15,000–₹20,000 per month

For 10k users:

👉 Cost per user ≈ ₹2/month

Very scalable.



MARKET ADVISORY

The system analyzes historical and real-time mandi price data to forecast short-term price trends for selected crops. Based on predicted price movement, storage feasibility, and crop perishability, the platform provides actionable hold or sell recommendations. This enables farmers to make informed marketing decisions and reduce losses due to price volatility.

Market Advisor

Select Crop

Soybean

Select Mandi

Nagpur

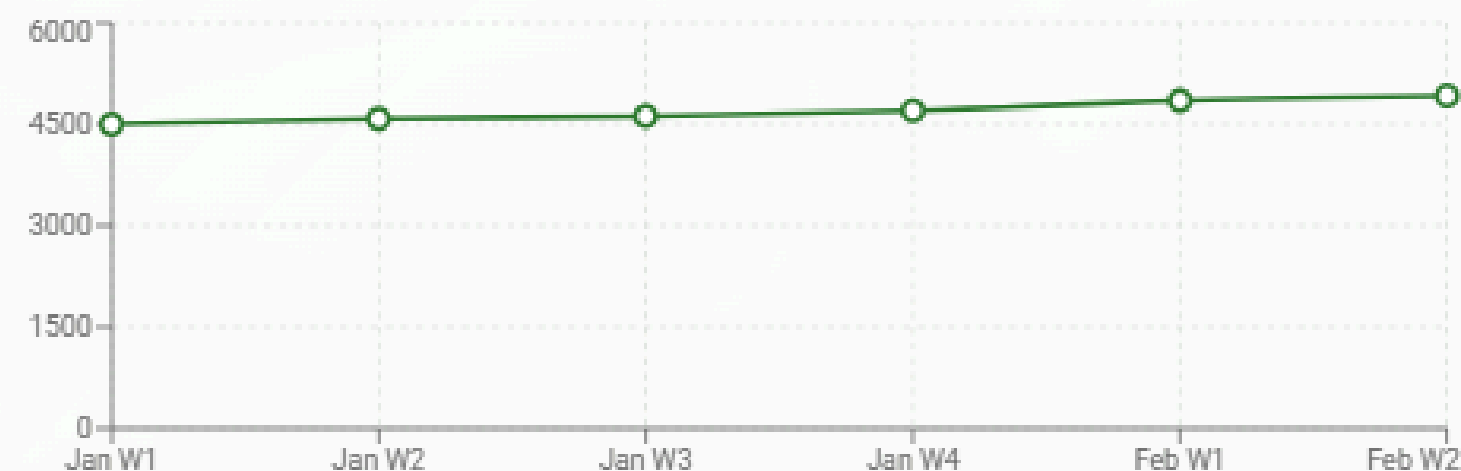
Predicted Price

Date	Predicted Price	Confidence	Recommendation
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2026-03-01	₹4980	55%	SELL
2026-03-05	₹4850	50%	HOLD

Why This Advice?

Soybean prices are expected to rise over the next 10 days due to increasing demand and limited supply. Selling around Feb 20 at ₹5180/quintal may be the best opportunity. After that, prices may stabilize or slightly decline.

Price Trend





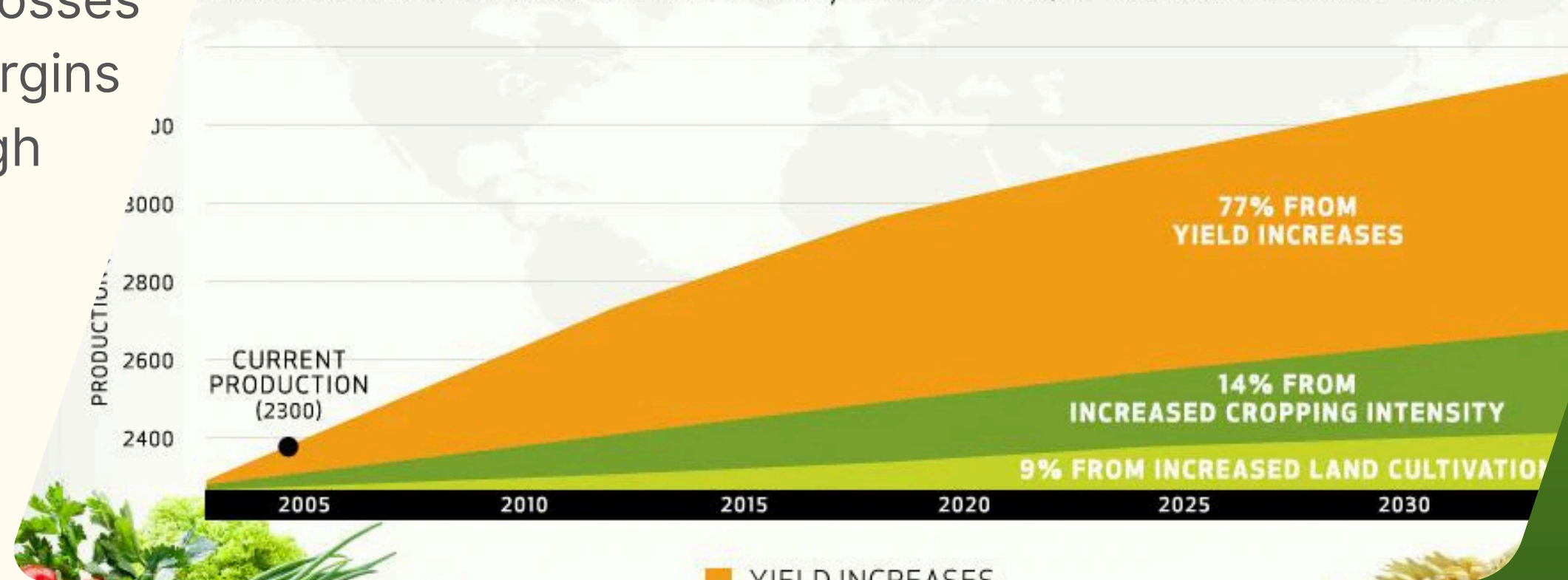
IMPACT METRICS



Expected Outcomes

- 10–20% increase in crop yield
- Up to 30% reduction in disease-related losses
- 15–25% improvement in farmer profit margins
- Reduced fertilizer and input costs through optimization

will need to come from increases in the yield achieved, or reductions in food waste.



VIDEO SOLUTION AND PROTOTYPE

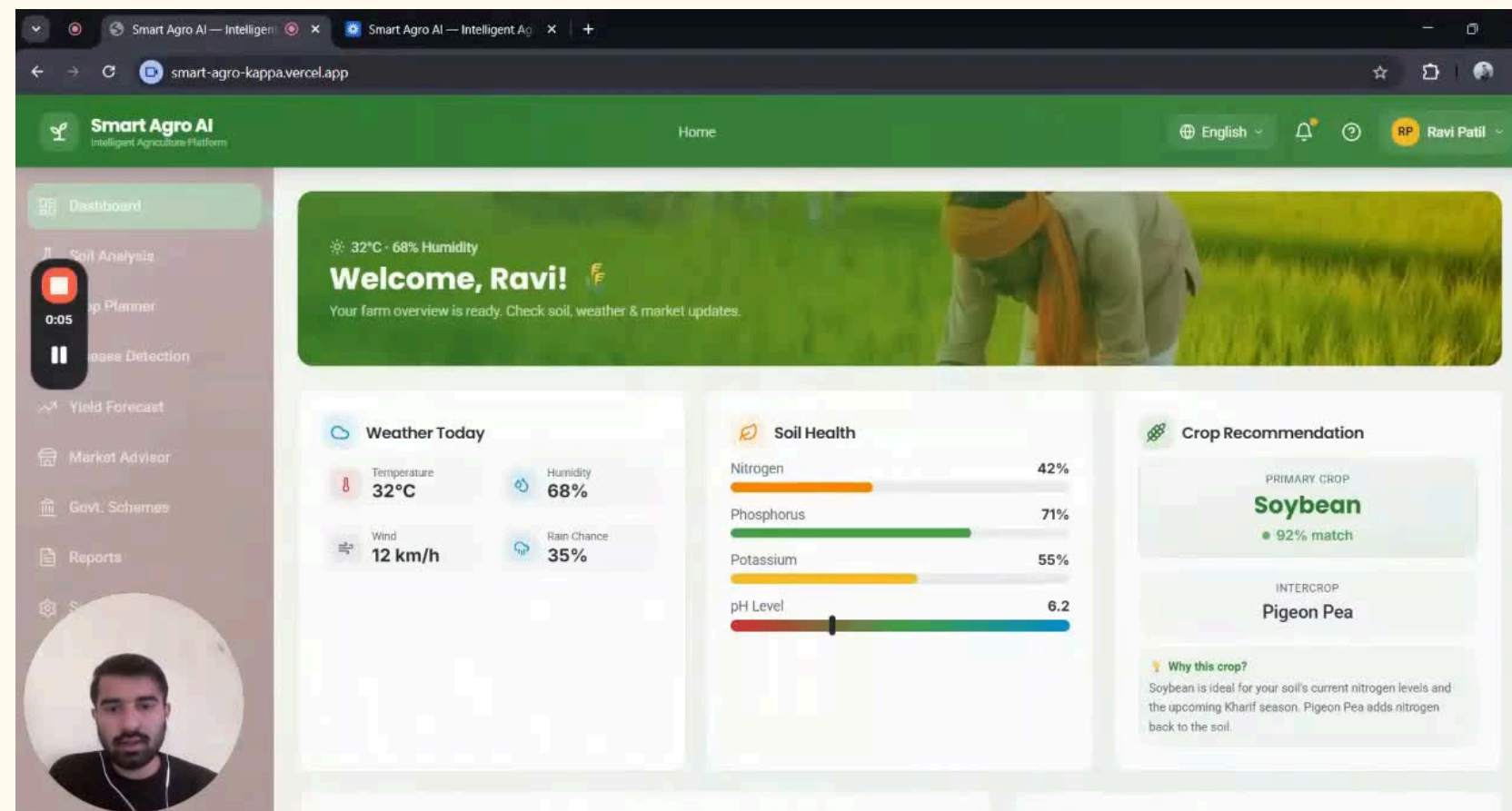


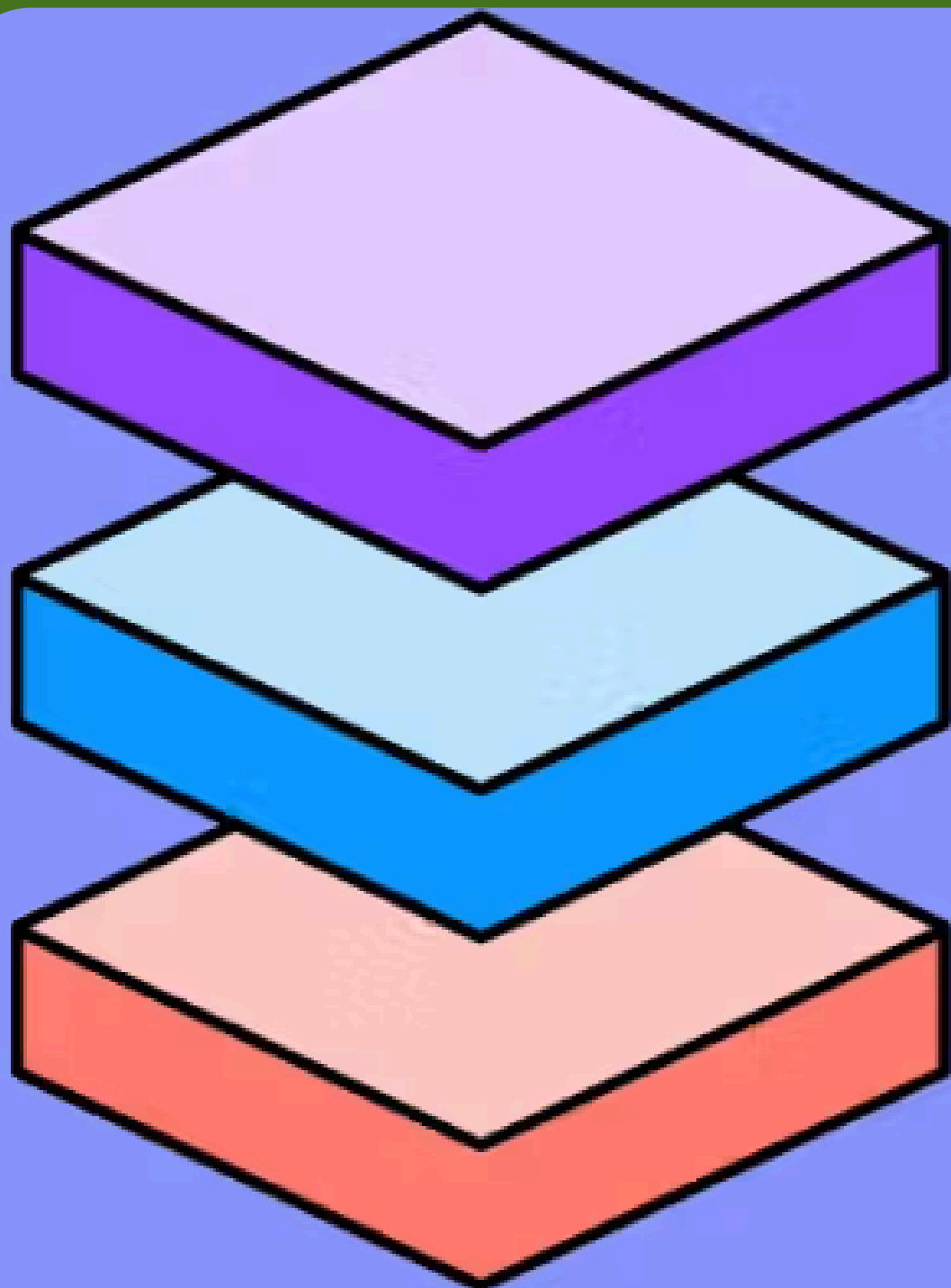
PROTOTYPE : https://smart-agro-git-master-yashp1395s-projects.vercel.app?vercel_share=iUWhYDNEji8NZx3Ckm9uHAXdgLVucSnw

VIDEO DESCRIPTION :

https://drive.google.com/file/d/1evxwtP3YhIhVex3uY5w2UpUigKbEBHJW/view?usp=drive_link

GITHUB REPO : <https://github.com/yashp1395/smart-agro.git>





TECH STACK (WHY IT'S SMART)

Frontend

- React.js for fast, scalable UI
- Recharts for clear data visualization

AI & ML

- CNN for disease image analysis
- LSTM for yield forecasting
- XGBoost for market price prediction

Backend & Infra

- Lovable Cloud as secure AI gateway
- REST APIs with rate limiting

Edge & Localization

- ESP32 for low-cost edge sensing
- English / Hindi (i18n) support



RESEARCH PAPER

- Advances in Soil Health Monitoring: A Review of Digital and Sensor-Based Tools
- Systematic Review Soil Nutrient Monitoring Technologies for Sustainable Agriculture: A Systematic Review
- Image-based crop disease detection using machine learning



WHY THIS SOLUTION STANDS OUT

- Solves real, on-ground agricultural problems
- Uses AI where it adds clear value
- Affordable for small and marginal farmers
- Scalable system architecture
- Provides actionable, decision-oriented outputs
- Strong end-to-end demo capability.

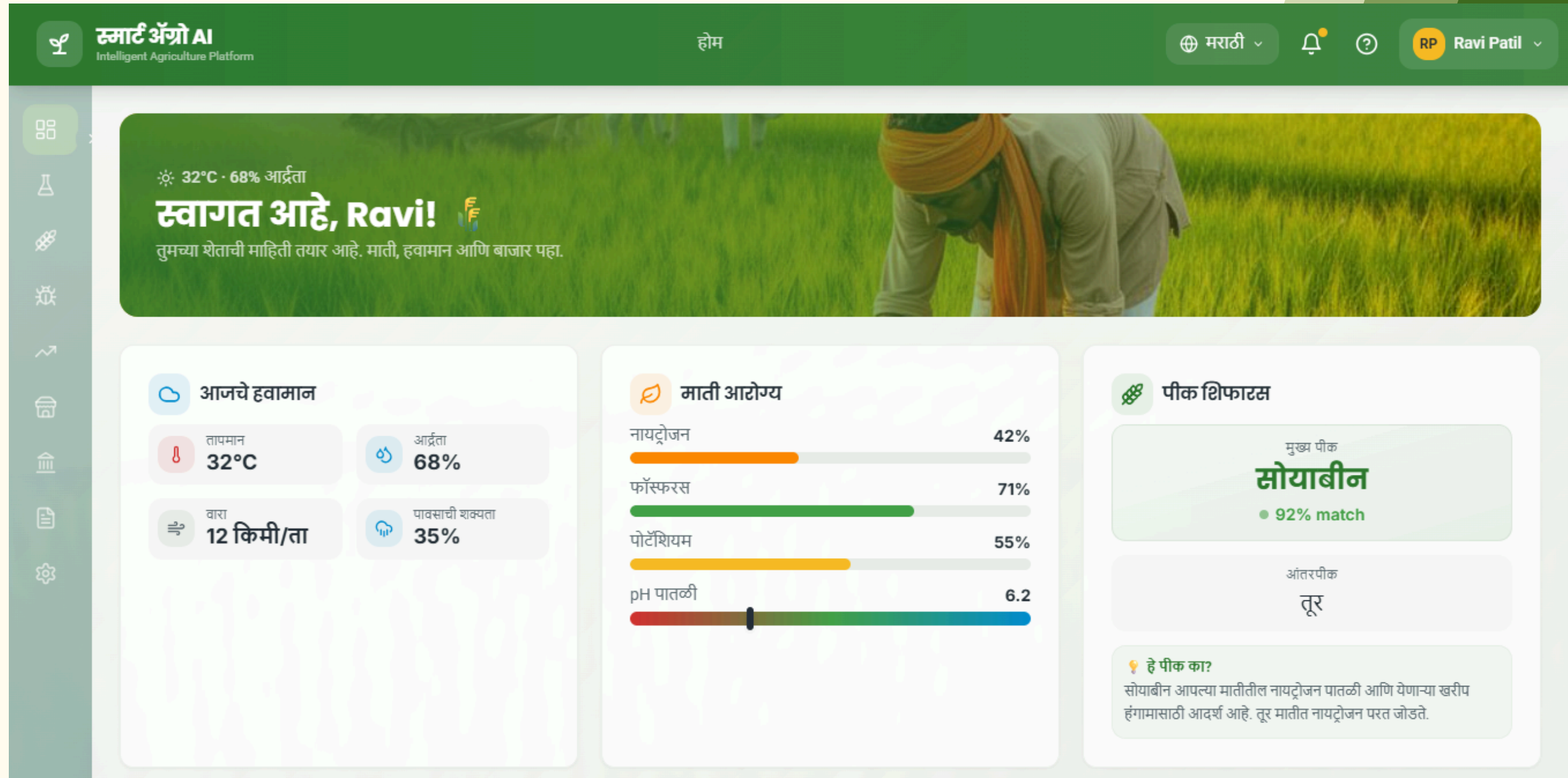




CONCLUSION



Smart Agro AI is a unified decision-support platform that combines soil intelligence, crop planning, disease detection, yield forecasting, and market advisory into a single, affordable system designed for real farmer needs.





THANK YOU